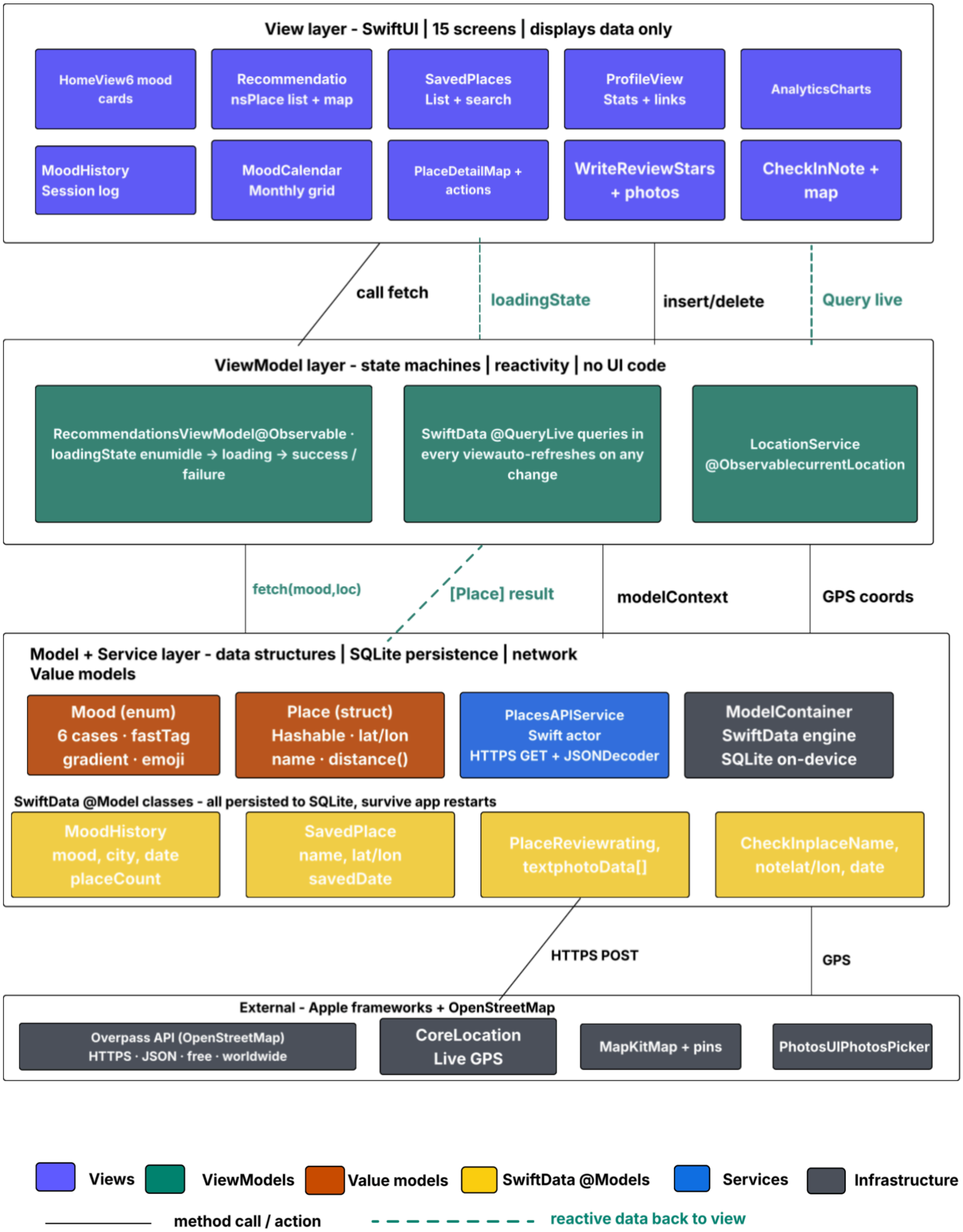
CSE 335 Class Project Phase III

a) Link for your technical presentation (Presentation length should not exceed 12 minutes and should not be less than 8 minutes).

https://youtu.be/tTgpcB9aYkA

b) Copy and Paste your Final MVMM Architecture diagram. This diagram should be complete and functionality of each model, view, view-model(or controller) need to explained clearly showing how each component interact with each other in your application



MVVM Architecture Diagram - Explanation

What it shows: The MVVM diagram shows how the app is structured internally across four layers. Data flows strictly downward through the layers, and reactive updates travel back upward automatically.

Layer 1 - View: This layer contains all 10 SwiftUI screens. The views are responsible only for displaying information. They never call the network directly and never write to the database themselves. All data they show comes from the ViewModel layer below.

Layer 2 - ViewModel: This is the brain of the app. RecommendationsViewModel manages a loading state that moves from idle to loading, then to either success or failure depending on the API response. SwiftData's Query acts as a live, reactive ViewModel for all stored data. Whenever anything is saved or deleted anywhere in the app, every screen that reads that data updates itself automatically. LocationService wraps the GPS and exposes the current coordinates as an observable property.

Layer 3 - Model and Service: This layer has two types of components. Value models are plain Swift types. Mood is an enum with six cases, and PlaceDetail is a struct. The four SwiftData Model classes, MoodHistory, SavedPlace, PlaceReview, and CheckIn, are the persistent data models stored in a SQLite database on the device. PlacesAPIService is a Swift actor that builds the geographic query and sends a live HTTPS request to the Overpass API, then decodes the JSON response into Place structs.

Layer 4 - External: This layer sits outside the app entirely. It includes the Overpass API, which provides real place data from OpenStreetMap, CoreLocation, which provides the live GPS position, MapKit, which renders the interactive map, and PhotosUI, which gives access to the camera roll when writing reviews.

How the arrows work: Solid arrows represent a method call or action flowing downward. Dashed green arrows represent reactive data flowing back up to the view automatically. No manual refresh is ever needed.

c) Upload your PPT slides to the Google Drive and share the link for your PPT Slides
https://drive.google.com/file/d/1Dpnwg8NUvcv4vD5rW6GX_7qFMF7W3z-E/view?usp=sharing